Programming in Python
IDC 212
Lecture - 9

October 11, 2025

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Numpy

Introduction to Tensor Shapes

- A **tensor** is a multi-dimensional array.
- Shape refers to the dimensions of the array.
- Axis indicates the direction along which operations like sum, mean, etc. are performed.

1D Array (Vector)

```python
import numpy as np
a = np.array([10, 20, 30])
print(a.shape)
```

Output: `(3,)`
Explanation: This is a 1D tensor with 3 elements. Only one axis (axis 0).

2D Array (Matrix)

```
b = np.array([[1, 2, 3],
              [4, 5, 6]])
print(b.shape)
```

Output: `(2, 3)`

Explanation: This is a 2D tensor with:

- Axis 0: 2 rows
- Axis 1: 3 columns

3D Tensor (Example)

```python
c = np.array([[[1, 2], [3, 4]],
              [[5, 6], [7, 8]]])
print(c.shape)
```

Output: `(2, 2, 2)`

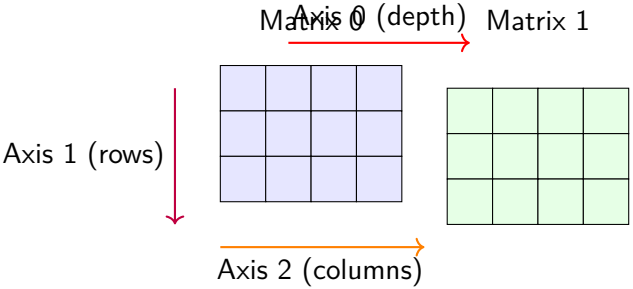
Explanation:

- Axis 0: 2 matrices
- Axis 1: 2 rows per matrix
- Axis 2: 2 columns per row

Visualizing 3D NumPy Array Shape: (2, 3, 4)

`np.zeros((2, 3, 4))` means:

- Axis 0 (depth): 2 matrices
- Axis 1 (rows): 3 rows per matrix
- Axis 2 (columns): 4 columns per row



Summary

- Shape defines the dimensions of the array.
- Axis 0 is usually the "rows", Axis 1 is "columns" in 2D.
- Axis increases with tensor rank: axis 2 for depth, etc.
- Understanding axes is key for operations like sum, mean, broadcasting.

Practice Question

What will be the output shape of the following array?

```
x = np.zeros((3, 4, 2))
```

- 3 blocks
- 4 rows per block
- 2 columns per row

Practice Question

What will be the output shape of the following array?

```
x = np.zeros((3, 4, 2))
```

Answer: `(3, 4, 2)`

- 3 blocks
- 4 rows per block
- 2 columns per row

Arithmetic Operations with Two Arrays

If we use another array instead of a scalar, the elements of both arrays will be component-wise combined:

```python
import numpy as np

A = np.array([[11, 12, 13],
              [21, 22, 23],
              [31, 32, 33]])
B = np.ones((3,3))

print("Adding to arrays: ")
print(A + B)

print("\nMultiplying two arrays: ")
print(A * (B + 1))
```

Output

Output:

```
Adding to arrays:
[[12. 13. 14.]
 [22. 23. 24.]
 [32. 33. 34.]]

Multiplying two arrays:
[[22. 24. 26.]
 [42. 44. 46.]
 [62. 64. 66.]]
```

Important Note

`A * B` in the previous example should not be mistaken for matrix multiplication. The elements are solely **component-wise multiplied**.

Matrix Multiplication

For this purpose, we can use the dot product. Using the previous arrays, we can calculate the matrix multiplication:

```
np.dot(A, B)
```

Output:

```
array([[36., 36., 36.],
       [66., 66., 66.],
       [96., 96., 96.]])
```

Definition of the Dot Product

The dot product is defined like this:
`np.dot(a, b, out=None)`

- For 2-D arrays: equivalent to matrix multiplication.
- For 1-D arrays: inner product of vectors.
- For scalars: simple multiplication.
- For N-D arrays: sum product over last axis of `a` and second-to-last of `b`.

Dot Product Parameters

Parameter	Meaning
`a`	array or array-like argument
`b`	array or array-like argument
`out`	optional parameter, must match the expected shape; improves performance

Examples of Using the Dot Product (1D & Scalars)

```python
print(np.dot(3, 4))        # 12

x = np.array([3])
y = np.array([4])
print(np.dot(x, y))        # 12

x = np.array([3, -2])
y = np.array([-4, 1])
print(np.dot(x, y))        # -14
```

Output:

```
12
12
-14
```

Examples of Using the Dot Product (2D Arrays)

```python
A = np.array([[1, 2, 3],
              [3, 2, 1]])
B = np.array([[2, 3, 4, -2],
              [1, -1, 2, 3],
              [1, 2, 3, 0]])

print(np.dot(A, B))
```

Output:

```
[[ 7  7 17  4]
 [ 9  9 19  0]]
```

Matrix Multiplication with np.dot

For two-dimensional arrays, `np.dot(A, B)` performs matrix multiplication.

```python
import numpy as np

A = np.array([[11, 12, 13],
              [21, 22, 23],
              [31, 32, 33]])
B = np.ones((3, 3))

R = np.dot(A, B)
print(R)
```

Output:

```
[[36. 36. 36.]
 [66. 66. 66.]
 [96. 96. 96.]]
```

Definition of the Dot Product

The dot product is defined as:
`np.dot(a, b, out=None)`

- Scalars: simple multiplication.
- 1D arrays: inner product.
- 2D arrays: matrix multiplication.
- N-D arrays: sum product over the last axis of `a` and the second-to-last of `b`.

Dot Product Parameters

Parameter	Meaning
`a`	array or array-like argument
`b`	array or array-like argument
`out`	optional parameter, must match expected shape; improves performance

Examples of Dot Product (Scalars & 1D)

```python
print(np.dot(3, 4))        # 12

x = np.array([3])
y = np.array([4])
print(np.dot(x, y))        # 12

x = np.array([3, -2])
y = np.array([-4, 1])
print(np.dot(x, y))        # -14
```

Output:

```
12
12
-14
```

Examples of Dot Product (2D Arrays)

```python
A = np.array([[1, 2, 3],
              [3, 2, 1]])
B = np.array([[2, 3, 4, -2],
              [1, -1, 2, 3],
              [1, 2, 3, 0]])

print(np.dot(A, B))
```

Output:

```
[[ 7  7 17  4]
 [ 9  9 19  0]]
```

Dot Product with 3D Arrays

```python
X = np.random.randint(1, 5, (3, 3, 3))
Y = np.random.randint(1, 5, (3, 3, 3))

R = np.dot(X, Y)

print("Shapes:")
print(X.shape, Y.shape, R.shape)
```

Output:

```
X.shape = (3, 3, 3)
Y.shape = (3, 3, 3)
R.shape = (3, 3, 3, 3)
```

The result is a 4D array due to contraction of axes.

Non-Symmetric 3D Dot Product

```python
X = np.random.randint(1, 5, (4, 2, 3))
Y = np.random.randint(1, 5, (2, 3, 5))

R = np.dot(X, Y)

print("Shapes:")
print(X.shape, Y.shape, R.shape)
```

Output:

```
X.shape = (4, 2, 3)
Y.shape = (2, 3, 5)
R.shape = (4, 2, 2, 5)
```

Arrays vs Matrices

Difference between 2D arrays and matrices:

```
A = np.array([[1, 2, 3],
              [2, 2, 2],
              [3, 3, 3]])
B = np.array([[ 3,  2,  1],
              [ 1,  2,  3],
              [-1, -2, -3]])

print(A * B)   # elementwise multiplication

MA = np.mat(A)
MB = np.mat(B)
print(MA * MB) # matrix multiplication
```

Note: Prefer `A @ B` in modern NumPy.

Logical Operators

We can apply logical operations elementwise on arrays using `np.logical_or` and `np.logical_and`:

```python
import numpy as np

a = np.array([[True, True],
              [False, False]])
b = np.array([[True, False],
              [True, False]])

print(np.logical_or(a, b))
print(np.logical_and(a, b))
```

Output:

```
[[ True  True]
 [ True False]]
[[ True False]
 [False False]]
```

Applying Operators on Arrays with Different Shapes

So far we have seen:

- Operator applied to an array and a scalar.
- Operator applied to two arrays of the same shape.

NumPy Broadcasting:

- If shapes differ, NumPy tries to stretch the smaller array.
- Rule: dimensions must match or one must be 1.
- Enables operations like adding vectors to matrices.

Example of Broadcasting

```
A = np.array([[1, 2, 3],
              [4, 5, 6]])
B = np.array([10, 20, 30])

print(A + B)
```

Output:

```
[[11 22 33]
 [14 25 36]]
```

Broadcasting

NumPy provides a powerful mechanism called **broadcasting**, which allows arithmetic operations on arrays of different shapes.

- The smaller array is virtually repeated to match the shape of the larger array.
- No actual copies are made, so broadcasting is memory efficient.
- Loops are executed internally in fast C code.

We demonstrate this with a simple example.

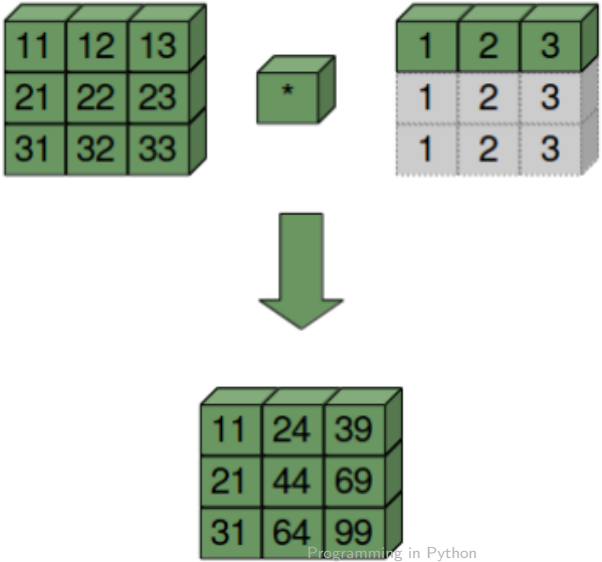
First Example of Broadcasting

```python
import numpy as np

A = np.array([[11, 12, 13],
              [21, 22, 23],
              [31, 32, 33]])
B = np.array([1, 2, 3])

print("Multiplication with broadcasting: ")
print(A * B)

print("Addition with broadcasting: ")
print(A + B)
```

Output:

```
Multiplication with broadcasting:
[[11 24 39]
 [21 44 69]
 [31 64 99]]
```

Visualization of Broadcasting



Second Example of Broadcasting

For this example, we first turn a row vector into a column vector:

```python
import numpy as np

B = np.array([1, 2, 3])
print(B[:, np.newaxis])
```

Output:

```
array([[1],
       [2],
       [3]])
```

Multiplication with Column Vector

Now we can multiply a matrix with this column vector using broadcasting:

```
A = np.array([[11, 12, 13],
              [21, 22, 23],
              [31, 32, 33]])

print(A * B[:, np.newaxis])
```

Output:

```
array([[11, 12, 13],
       [42, 44, 46],
       [93, 96, 99]])
```

Explanation of Broadcasting (Second Example)

Here `B[:, np.newaxis]` is treated as if it were:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$$

This expanded form allows elementwise multiplication with `A`.

Third Example of Broadcasting

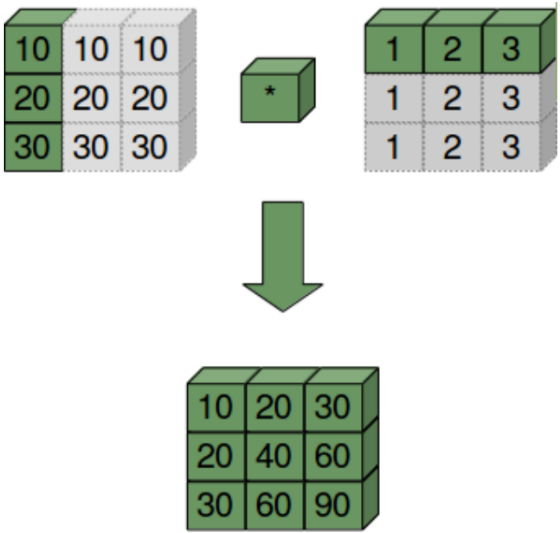
We can also use broadcasting with 1D arrays turned into column vectors:

```
A = np.array([10, 20, 30])
B = np.array([1, 2, 3])

print(A[:, np.newaxis])
```

Output:

```
array([[10],
       [20],
       [30]])
```

Visualization of Broadcasting



Flatten in NumPy

The method `flatten()` returns a copy of the array collapsed into one dimension. It accepts an optional keyword `order`, which can take values:

- `"C"` (default): C-style row-major order (last index varies fastest).
- `"F"`: Fortran-style column-major order (first index varies fastest).
- `"A"`: Preserve the array's original order (C/Fortran).

Example of Flatten

```python
import numpy as np

A = np.array([[[ 0,  1],
               [ 2,  3],
               [ 4,  5],
               [ 6,  7],
               [ 8,  9],
               [10, 11],
               [12, 13],
               [14, 15],
               [16, 17],
               [18, 19],
               [20, 21],
               [22, 23]]])

Flattened_X = A.flatten()
print(Flattened_X)
```

Output (default order "C"):

\scriptsize

Flatten with Different Orders

```python
print(A.flatten(order="C"))
print(A.flatten(order="F"))
print(A.flatten(order="A"))
```

Outputs:

```
\scriptsize
C order:
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23]

F order:
[ 0  8 16  2 10 18  4 12 20  6 14 22  1  9 17  3 11 19  5 13 21  7 15 23]

A order:
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23]
```

Notes on Flatten

- `flatten()` always returns a **copy** of the data.
- Compare with `ravel()`, which may return a **view** when possible.
- Choosing the right order (`"C"` or `"F"`) depends on how the array is stored or expected in memory.

Ravel in NumPy

The method `ravel()` returns a flattened one-dimensional array.

- Unlike `flatten()`, it returns a **view** of the array whenever possible.
- A copy is made only if needed.
- The order of flattening can be specified with the keyword `order`.

Available orders:

- `"C"`: C-style row-major order (last index varies fastest).
- `"F"`: Fortran-style column-major order (first index varies fastest).
- `"A"`: Use Fortran order if array is Fortran-contiguous, else C-order.
- `"K"`: Keep the order as stored in memory (except reversed if strides are negative).

Example of Ravel

```python
import numpy as np

A = np.array([[[ 0,  1],
               [ 2,  3],
               [ 4,  5],
               [ 6,  7],
               [ 8,  9],
               [10, 11],
               [12, 13],
               [14, 15],
               [16, 17],
               [18, 19],
               [20, 21],
               [22, 23]]])

print(A.ravel())
print(A.ravel(order="A"))
print(A.ravel(order="F"))
print(A.ravel(order="K"))
```

Output of Ravel

Outputs:

```
\scriptsize
C order (default):
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23]

A order:
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23]

F order:
[ 0  8 16  2 10 18  4 12 20  6 14 22  1  9 17  3 11 19  5 13 21  7 15 23]

K order:
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23]
```

Notes: Flatten vs Ravel

- `flatten()` always returns a **copy**.
- `ravel()` returns a **view** when possible (more memory efficient).
- Both support `order="C"`, `"F"`, `"A"`, and `"K"`.
- Prefer `ravel()` when you only need a temporary 1D view of data.

Vector Stacking (Row vs Column Stack)

```python
import numpy as np

A = np.array([3, 4, 5])
B = np.array([1, 9, 0])

print(np.row_stack((A, B)))
print(np.column_stack((A, B)))
print(A.shape)
```

Output:

```
[[3 4 5]
 [1 9 0]]

[[3 1]
 [4 9]
 [5 0]]
```

Vector Stacking (Column Stack Multiple)

```
A = np.array([[3, 4, 5],
              [1, 9, 0],
              [4, 6, 8]])

print(np.column_stack((A, A, A)))
```

Output:

```
[[3, 4, 5, 3, 4, 5, 3, 4, 5],
 [1, 9, 0, 1, 9, 0, 1, 9, 0],
 [4, 6, 8, 4, 6, 8, 4, 6, 8]]
```

Vector Stacking (Column Stack with Rows)

```python
print(np.column_stack((A[0], A[0], A[0])))
```

Output:

```
[[3 3 3]
 [4 4 4]
 [5 5 5]]
```

Vector Stacking (Depth Stack)

```python
print(np.dstack((A, A, A)))
```

Output:

```
array([[[3, 3, 3],
        [4, 4, 4],
        [5, 5, 5]],

       [[1, 1, 1],
        [9, 9, 9],
        [0, 0, 0]],

       [[4, 4, 4],
        [6, 6, 6],
        [8, 8, 8]]])
```

Vector Norms

Definition: For $x \in \mathbb{R}^n$,

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|, \quad \|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}, \quad \|x\|_\infty = \max_i |x_i|$$

Vector Norms in Python

Python Example:

```python
import numpy as np
x = np.array([1, -2, 3])

# Vector norms
np.linalg.norm(x, 1)       # 1-norm
np.linalg.norm(x, 2)       # 2-norm
np.linalg.norm(x, np.inf) # Infinity-norm
```

Matrix Norms

Definition: For $A \in \mathbb{R}^{m \times n}$,

- $\|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2}$ (Frobenius norm)
- $\|A\|_1 = \max_j \sum_i |a_{ij}|$ (max column sum)
- $\|A\|_\infty = \max_i \sum_j |a_{ij}|$ (max row sum)
- $\|A\|_2 =$ largest singular value of A (spectral norm)

Matrix Norms in Python

Python Example:

```python
import numpy as np
A = np.array([[1, 2],
              [3, 4]])

np.linalg.norm(A, 'fro')  # Frobenius norm
np.linalg.norm(A, 1)      # 1-norm (max column sum)
np.linalg.norm(A, np.inf) # Infinity-norm (max row sum)
np.linalg.norm(A, 2)      # Spectral norm (largest singular
    value)
```

Important Remark

- For vectors:
$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$
is the Euclidean length.
- For matrices:
$$\|A\|_2 = \sigma_{\max}(A)$$
where $\sigma_{\max}(A)$ is the largest singular value.
- Hence, **matrix 2-norm = spectral norm**.

Objective Question 1: Vector Norm

For $x = (3, -4)$, compute $\|x\|_2$.

A. 5

B. 7

C. 12

D. 25

Objective Question 1: Vector Norm

For $x = (3, -4)$, compute $\|x\|_2$.

A. 5

B. 7

C. 12

D. 25

Solution:

$$\|x\|_2 = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Correct answer: **A**

Objective Question 2: Infinity Norm

For $x = (2, -7, 5)$, compute $\|x\|_\infty$.

A. 2

B. 5

C. 7

D. 14

Objective Question 2: Infinity Norm

For $x = (2, -7, 5)$, compute $\|x\|_\infty$.

A. 2

B. 5

C. 7

D. 14

Solution:

$$\|x\|_\infty = \max(|2|, |-7|, |5|) = 7$$

Correct answer: **C**

Objective Question 3: Matrix 1-Norm

Let

$$A = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}$$

Compute $\|A\|_1$.

A. 4

B. 6

C. 7

D. 10

Objective Question 3: Matrix 1-Norm

Let

$$A = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}$$

Compute $\|A\|_1$.

A. 4

B. 6

C. 7

D. 10

Solution:

$$\text{Col 1 sum} = |1| + |3| = 4, \quad \text{Col 2 sum} = |-2| + |4| = 6$$

$$\|A\|_1 = \max(4, 6) = 6$$

Correct answer: **B**

Objective Question 4: Matrix Infinity Norm

Let

$$B = \begin{pmatrix} 2 & -3 \\ -1 & 4 \end{pmatrix}$$

Compute $\|B\|_\infty$.

A. 5

B. 6

C. 7

D. 8

Objective Question 4: Matrix Infinity Norm

Let

$$B = \begin{pmatrix} 2 & -3 \\ -1 & 4 \end{pmatrix}$$

Compute $\|B\|_\infty$.

A. 5

B. 6

C. 7

D. 8

Solution:

$$\text{Row 1 sum} = |2| + |-3| = 5, \quad \text{Row 2 sum} = |-1| + |4| = 5$$

$$\|B\|_\infty = \max(5, 5) = 5$$

Correct answer: **A**

Objective Question 5: Frobenius Norm

For

$$C = \begin{pmatrix} 1 & 2 \\ -2 & 3 \end{pmatrix}$$

compute $\|C\|_F$.

A. $\sqrt{10}$

B. $\sqrt{15}$

C. $\sqrt{18}$

D. $\sqrt{20}$

Objective Question 5: Frobenius Norm

For

$$C = \begin{pmatrix} 1 & 2 \\ -2 & 3 \end{pmatrix}$$

compute $\|C\|_F$.

A. $\sqrt{10}$
B. $\sqrt{15}$
C. $\sqrt{18}$
D. $\sqrt{20}$

Solution:

$$\|C\|_F = \sqrt{1^2 + 2^2 + (-2)^2 + 3^2} = \sqrt{1 + 4 + 4 + 9} = \sqrt{18}$$

Correct answer: **C**

Determinant of a Square Matrix

```python
import numpy as np

# Define a 2x2 matrix
A = np.array([[4, 2],
              [3, 1]])

det_A = np.linalg.det(A)
print("Determinant of A:", det_A)
```

Output:

```
Determinant of A: -2.0
```

The determinant exists and is non-zero, so the matrix is invertible.

Case: Determinant = 0 (Singular Matrix)

```python
# Define a singular matrix
# (rows are linearly dependent)
B = np.array([[2, 4],
              [1, 2]])

det_B = np.linalg.det(B)
print("Determinant of B:", det_B)
```

Output:

```
Determinant of B: 0.0
```

The determinant is zero, so the matrix is singular and has no inverse.

Case: Determinant Not Defined (Non-Square Matrix)

```python
# Non-square matrix
C = np.array([[1, 2, 3],
              [4, 5, 6]])

# This will raise an error
det_C = np.linalg.det(C)
```

Error:

```
LinAlgError: Last 2 dimensions of the array
must be square
```

For non-square matrices, the determinant is not defined.

Inverse of an Invertible Matrix

```python
import numpy as np

A = np.array([[4, 7],
              [2, 6]])

det_A = np.linalg.det(A)
print("Determinant:", det_A)

A_inv = np.linalg.inv(A)
print("Inverse:\n", A_inv)

# Verify A * A_inv = I
print("Check:\n", np.dot(A, A_inv))
```

Output:

```
Determinant: 10.0
Inverse:
```

Case: Singular Matrix (No Inverse)

```
B = np.array([[2, 4],
              [1, 2]])

det_B = np.linalg.det(B)
print("Determinant:", det_B)

B_inv = np.linalg.inv(B)
```

Output:

```
Determinant: 0.0
LinAlgError: Singular matrix
```

Here $\det(B) = 0$, so B^{-1} does not exist.

Case: Non-Square Matrix (Inverse Undefined)

```
C = np.array([[1, 2, 3],
              [4, 5, 6]])

np.linalg.inv(C)
```

Error:

```
LinAlgError: Last 2 dimensions of the array
must be square
```

For non-square matrices, the inverse is not defined.

Matrix Decompositions in NumPy

NumPy provides several important matrix decompositions:

- **QR Decomposition**: $A = QR$
- **Eigen Decomposition**: $Av = \lambda v$
- **Singular Value Decomposition (SVD)**: $A = U\Sigma V^T$
- **Cholesky Decomposition**: $A = LL^T$

These are fundamental tools in linear algebra, data science, and machine learning.

QR Decomposition

Theory:

$$A = QR, \quad Q^T Q = I, \; R \text{ is upper triangular}$$

Code:

```python
import numpy as np

A = np.array([[1, 1, 0],
              [1, 0, 1],
              [0, 1, 1]])

Q, R = np.linalg.qr(A)
print("Q:\n", Q)
print("R:\n", R)
print("Check Q*R:\n", Q @ R)
```

Eigen Decomposition

Theory:

$$Av = \lambda v, \quad A = V\Lambda V^{-1}$$

Code:

```
vals, vecs = np.linalg.eig(A)
print("Eigenvalues:", vals)
print("Eigenvectors:\n", vecs)

# Verify Av = lambda*v for first eigenpair
print("Check:", A @ vecs[:,0],
                vals[0] * vecs[:,0])
```

Singular Value Decomposition (SVD)

Theory:

$$A = U\Sigma V^T$$

Code:

```python
U, S, Vt = np.linalg.svd(A)

print("U:\n", U)
print("Singular values:", S)
print("V^T:\n", Vt)

# Reconstruct A
Sigma = np.zeros_like(A, dtype=float)
np.fill_diagonal(Sigma, S)
print("Reconstructed:\n", U @ Sigma @ Vt)
```

Cholesky Decomposition

Theory: For symmetric positive-definite A:

$$A = LL^T, \quad L \text{ is lower triangular}$$

Code:

```
A = np.array([[4, 2],
              [2, 3]])

L = np.linalg.cholesky(A)
print("L:\n", L)
print("Check:\n", L @ L.T)
```

Summary of Decompositions

Decomposition	Form	Applications
QR	$A = QR$	Least squares, orthogonalization
Eigen	$A = V\Lambda V^{-1}$	PCA, control, spectral analysis
SVD	$A = U\Sigma V^T$	PCA, compression, pseudoinverse
Cholesky	$A = LL^T$	Optimization, covariance

Eigenvalues and Eigenvectors: Definition

For a square matrix $A \in \mathbb{R}^{n \times n}$:

$$Av = \lambda v$$

- v: eigenvector (non-zero vector)
- λ: eigenvalue (scalar)

Example:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Av = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2v$$

Thus, $\lambda = 2$, $v = (1, 0)^T$.

Geometric Meaning

- Eigenvectors are **directions** preserved under transformation.
- Eigenvalues are the **stretching/shrinking factors**.
- $\lambda = 0$: vector is flattened (singular matrix).
- Complex eigenvalues: describe **rotations**.

Illustration:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is a rotation by 90°. Its eigenvalues are $\pm i$.

Eigenfunctions (Operators on Functions)

Generalization from matrices to **operators** T:

$$Tf = \lambda f$$

Example: Derivative Operator

$$T = \frac{d}{dx}, \quad \frac{df}{dx} = \lambda f(x)$$

Solution:

$$f(x) = Ce^{\lambda x}$$

So exponential functions are eigenfunctions of $\frac{d}{dx}$ with eigenvalue λ.

Eigenfunctions: Laplacian Example

$$\nabla^2 f = \lambda f$$

Solutions: sines and cosines are eigenfunctions of the Laplacian. This is the foundation of Fourier series and PDE solutions.

In quantum mechanics: Schrödinger equation is an eigenvalue problem: Hamiltonian operator $\hat{H}\psi = E\psi$

- ψ: eigenfunction (wavefunction)
- E: eigenvalue (energy)

Applications of Eigenvalues/Eigenfunctions

- **Linear Algebra**: Principal Component Analysis (PCA).
- **Differential Equations**: Vibrations, heat conduction.
- **Quantum Mechanics**: Operators, wavefunctions, energy levels.
- **Graph Theory**: Spectral clustering, network connectivity.
- **Data Science**: Dimensionality reduction, embeddings.

Eigenvalues in NumPy

```python
import numpy as np

A = np.array([[2, 0],
              [0, 3]])

# Eigen decomposition
vals, vecs = np.linalg.eig(A)
print("Eigenvalues:", vals)
print("Eigenvectors:\n", vecs)

# Verify Av = lambda*v
print("Check:", A @ vecs[:,0],
              vals[0] * vecs[:,0])
```

Eigenvectors vs Eigenfunctions

Eigenvectors (finite-dimensional):

$$Av = \lambda v$$

- A: square matrix ($n \times n$) - v: eigenvector (finite list of numbers) - λ: eigenvalue (scalar)

Eigenfunctions (infinite-dimensional):

$$Tf(x) = \lambda f(x)$$

- T: linear operator (e.g., derivative, Laplacian) - $f(x)$: eigenfunction (function, not just vector) - λ: eigenvalue
Eigenfunctions are the *function-space analog* of eigenvectors.

Example: Derivative Operator

Operator:

$$T = \frac{d}{dx}, \quad \frac{df}{dx} = \lambda f(x)$$

Solve differential equation:

$$f(x) = Ce^{\lambda x}$$

- Exponential functions are eigenfunctions of $\frac{d}{dx}$.
- Eigenvalue is λ.

Example: Laplacian Operator

Consider the 1D Laplacian:

$$-\frac{d^2 f}{dx^2} = \lambda f(x), \quad f(0) = f(L) = 0$$

Solutions:

$$f_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, 3, \ldots$$

- Eigenfunctions: sine functions.
- Eigenvalues: squares of integer multiples of π/L.
- Basis for Fourier series and PDE solutions.

How to Approximate Eigenfunctions in NumPy

- Operators are infinite-dimensional $\Rightarrow$ cannot directly compute in NumPy.
- Idea: **discretize the operator** using finite differences.
- Example: approximate second derivative by a matrix (discretized Laplacian).
- Then compute eigenvalues/eigenvectors of this matrix.
- The eigenvectors approximate the continuous eigenfunctions.

NumPy Example: Discretized Laplacian

```python
import numpy as np
import matplotlib.pyplot as plt
# Discretize [0,1] into n interior points
n = 50
h = 1.0 / (n+1)
# Finite-difference Laplacian matrix
L = -2*np.eye(n) + np.eye(n,1) + np.eye(n,-1)
L = L / (h**2)
# Eigen decomposition
vals, vecs = np.linalg.eigh(L)
print("First 3 eigenvalues:", vals[:3])
# Plot first 3 eigenfunctions (discretized)
x = np.linspace(0,1,n+2)[1:-1]
plt.plot(x, vecs[:,0], label="1st mode")
plt.plot(x, vecs[:,1], label="2nd mode")
plt.plot(x, vecs[:,2], label="3rd mode")
plt.legend(); plt.show()
```

Output and Interpretation

- Eigenvalues approximate $\pi^2, (2\pi)^2, (3\pi)^2, \ldots$.
- Eigenvectors approximate sine functions $\sin(n\pi x)$.
- Plot shows sinusoidal eigenmodes:
  - 1st eigenfunction: one half-wave (fundamental mode).
  - 2nd eigenfunction: one full sine wave.
  - 3rd eigenfunction: one and a half sine waves, etc.
- These discrete eigenvectors are the **numerical approximation** of eigenfunctions.

Thank You